



A dual-mode MRAM-TCAM with intrinsic PUF for energy efficiency and data security[☆]

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ABSTRACT

Spin-transfer torque magnetic random-access memory (STT-MRAM)-based non-volatile ternary content-addressable memory (TCAM) is a promising solution for network acceleration and edge devices, featuring near-zero static power consumption and high integration density. However, its practical deployment is hindered by high power consumption in dynamic search operations and the lack of intrinsic hardware authentication. This paper presents a dual-mode MRAM-TCAM design, where a low-power 2-bit encoding scheme significantly mitigates match-line discharge activity and dynamic search energy. An intrinsic physical unclonable function (PUF) is also integrated to extract device fingerprints without disrupting TCAM storage operations, enabling built-in identity authentication. Simulation results based on 65 nm technology demonstrate the proposed design achieves 1.08 fJ/bit search energy and 0.95 ns search latency. The PUF exhibits a worst-case bit error rate (BER) of 1.17% at 120 °C and 0.29% at 1.4 V supply voltage, and its responses pass the NIST SP 800-22 test suite, validating the design's significant improvements in energy efficiency and security.

1. Introduction

Ternary content-addressable memory (TCAM) is a key hardware accelerator for high-speed routing lookup, network intrusion detection, and real-time data processing due to its ability to perform parallel search and matching within a single cycle [1–3]. With the widespread adoption of 5G communication and edge computing, the demand for TCAM continues to grow. Conventional static random-access memory (SRAM)-based TCAMs suffer from high static power consumption and limited storage density due to their volatile nature. Recently, spin-transfer torque magnetic random-access memory (STT-MRAM)-based non-volatile TCAMs have emerged as a promising alternative, offering near-zero static power and high integration [4–6].

Despite these advantages, the large-scale implementation of MRAM-TCAM faces two major challenges. First, power consumption is exacerbated by factors such as high write current, array parasitic capacitance, redundant circuits, static leakage, and match-line charging/discharging during parallel searches. This is particularly critical in resource-constrained Internet of Things (IoT) devices [7], making power efficiency a primary bottleneck. Zhu et al. proposed a

Spin-Orbit Torque MRAM-TCAM with a 6T-4MTJ cell and a cross-layer three-level reliability amortizing scheme [8]. This design decouples read/write paths to enhance search robustness, achieving an 8.5× larger match-line voltage swing (for better sensing reliability) and 3 fJ per search operation. Dhull et al. introduced an energy-efficient differential spin hall effect (DSHE)-based ternary content-addressable memory with a compact 3T-2MTJ cell [9]. This approach separates search/write current paths, achieving 30% higher search energy efficiency, 47.5% higher write efficiency, and 42.1% smaller layout area than a spin-orbit torque magnetic random-access memory (SOT-MRAM). Zhou et al. designed a high-sensing-margin 6T-2MTJ SOT-MRAM-TCAM for energy-efficient similarity search in memory-augmented neural networks (MANNs) [10]. Its cell structure boosts critical sensing margin by 2–3×, enabling more reliable parallel searches, and supports 4–8× enhanced computational parallelism while maintaining superior search energy over other contemporary non-volatile TCAM technologies. Garzon et al. presented “AM4”, a reconfigurable STT-MTJ crossbar for in-memory computing [11]. This versatile array can be configured as content-addressable memory (CAM), TCAM, ACAM, or associative processor, addressing the von Neumann data

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movement bottleneck. It achieves 10× performance and 60× energy efficiency gains over conventional designs.

Additionally, existing MRAM-TCAM designs lack native hardware security mechanisms, making them susceptible to physical side-channel attacks, data theft, and chip cloning. Moreover, TCAMs lack authentication during routing lookup, allowing attackers to forge malicious routing entries with shorter prefixes to redirect traffic to controlled nodes, leading to packet tampering or dropping [12]. Current protection measures include data encryption, physical shielding, and volatile key storage. Dadzie et al. proposed the non-volatile memory (NVM)-Shelf hybrid encryption framework, which divides data into active and inactive blocks, using modified counter mode (CTR) and electronic codebook (ECB) modes with key stream reuse, enabling low-flip encryption for all NVM segments in ScratchPad Memory (SPM) systems. However, it introduces up to 41% runtime overhead in the kernel [13]. Watanabe et al. designed a non-oriented silicon steel U-shaped magnetic shield, but the open design risks lateral field penetration and micro-probes can cause magnetic domain flipping through gaps [14]. Siddiqi et al. implemented GIFT-128 using a 40 nm resistive random-access memory (RRAM) 1T1R structure, compressing one encryption round into a single read, achieving a low-power side-channel resistant scheme with 242 pJ and 0.0034 mm². However, it is limited by write endurance, XOR efficiency, and scalability [15].

In parallel, physical unclonable functions (PUFs) based on emerging non-volatile memories have demonstrated exceptional reliability and energy efficiency as hardware security primitives [16]. Recent works further advance PUF designs: an MRAM PUF with offset-canceling sense amplifier and automatic write-back attains ideal uniformity and inter-Hamming distance [17]; a unified TRNG and PUF exploits RRAM switching time to achieve zero error after recombination [18]; a dynamic challenge cross-selection MRAM PUF resists machine learning attacks with nearly 50% attack accuracy and 9.975 fJ/bit energy [19]; and a dual-mode SRAM/inverter PUF reduces BER to 0.0053% over a wide voltage-temperature range [20]. Existing PUF designs typically operate as discrete hardware modules, which introduce area and power overhead from additional inter-module data buses. Integrating PUF functionality directly into content-addressable memory can eliminate this overhead by reusing the intrinsic process variations of the storage cells themselves.

This paper proposes a dual-mode MRAM-TCAM with intrinsic PUF design. The core innovations are a low-power 2-bit encoding scheme that reduces dynamic search energy and an intrinsic PUF for hardware authentication, without compromising TCAM functionality. The main contributions are:

- (a) **Low-power search:** Reduced match-line activation probability by 50% and dynamic search energy by 60.3% compared to conventional schemes. This 2-bit encoding cuts match-line discharge with no extra latency/area.
- (b) **Intrinsic PUF:** Hardware overhead of only 267 transistors (5.21% of storage module) while maintaining data integrity. The PUF leverages MRAM leakage variations for unique fingerprints, with no TCAM interference.
- (c) **Simulation validation:** Search energy reduced to 39.7% of conventional designs; PUF bit error rate (BER) of 1.17% at 120 °C and 0.29% at 1.4 V, passing NIST SP 800-22 tests. Results confirm balanced energy/security gains with retained TCAM high-speed search.

2. Dual-mode MRAM-TCAM with intrinsic PUF design

2.1. Design overview

The dual-mode MRAM-TCAM PUF design integrates ternary content-addressable memory functionality with physical unclonable function capabilities. The system supports two operational modes: in TCAM mode, the system performs conventional ternary content search and matching operations; in PUF mode, the system leverages inherent process variations from the manufacturing process to generate secure responses. Both modes achieve seamless switching through external control signals while sharing identical physical memory cell resources.

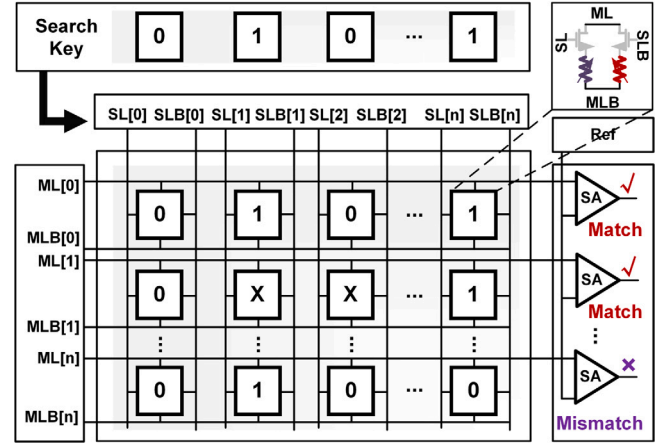


Fig. 1. Traditional MRAM-TCAM design.

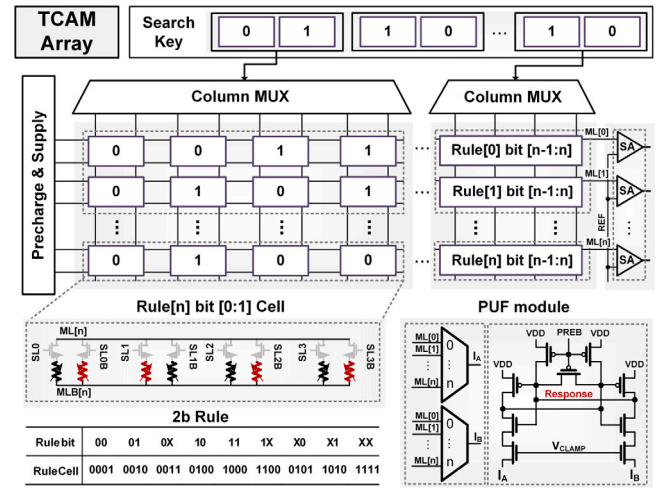


Fig. 2. Proposed 2-bit encoded MRAM-TCAM design.

2.2. 2-Bit encoded MRAM-TCAM strategy

As shown in Fig. 1, conventional MRAM-TCAM employs a parallel comparison search architecture, where the storage module is organized into an $N \times N$ array of 2T-2magnetic tunnel junction (MTJ) cells. During search operations, all match lines (MLs) are precharged. Subsequently, the search key is broadcast to all columns via search lines (SLs). Any cell mismatch pulls down the ML through a low-resistance path, causing the total capacitance driven per search to scale linearly with the number of entries, thereby dominating dynamic energy consumption. Although STT-MRAM exhibits significantly lower static leakage compared to SRAM, the cumulative standby power remains non-negligible at high entry depths typical of edge computing scenarios.

To address these limitations, this paper proposes a TCAM design based on a 2-bit encoding scheme. As shown in Fig. 2, the design partitions k -bit storage content (composed of {0, 1, x}) into 2-bit segments. Each segment utilizes a compact 2-bit to 4-bit encoding mapping: 2-bit patterns (where 'x' denotes a wildcard) are mapped to a 4-bit mask. The four bits of this mask correspond to binary tuples (00, 01, 10, 11); if the pattern covers a specific tuple, the corresponding mask bit is set to '1'. For instance, pattern '00' is encoded as '1000', while '1x' is encoded as '0011'. During search operations, the input search key is similarly encoded. These encoded signals activate specific storage cells for comparison. The 2-bit encoding scheme maps the input data into a 4-bit mask to implement a 1-of-4 selection mechanism. While a

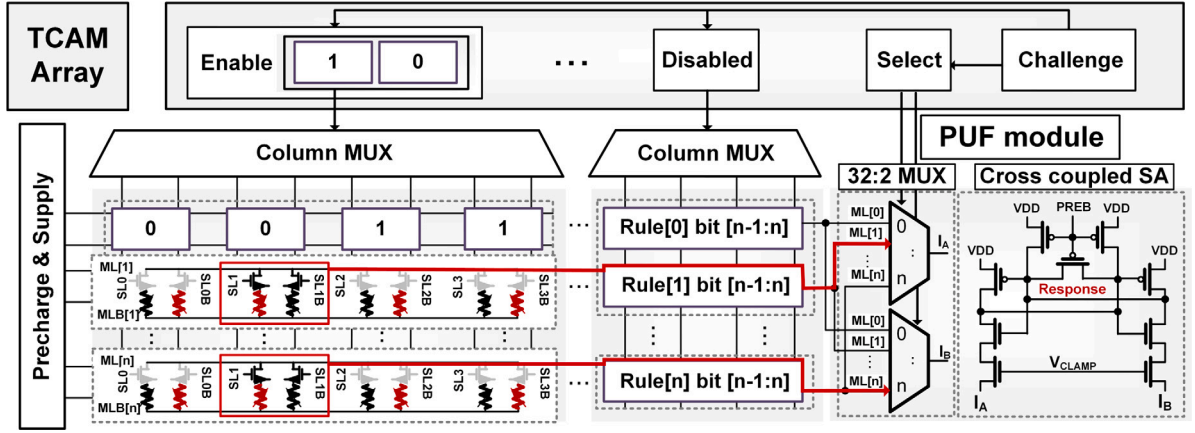


Fig. 3. Proposed PUF design: multiplexers select cells by challenge, cross-coupled SA compares discharge currents to generate response.

conventional search of 2-bit data involves two active comparison units, the proposed mapping ensures that only one discharge path is activated within each 2-bit comparison block. This reduces the theoretical match-line activation probability by 50%, thereby substantially mitigating the dynamic search power consumption.

2.3. Dual-mode reconfigurable MRAM-TCAM PUF

This paper proposes an intrinsic MRAM-TCAM PUF design, as shown in Fig. 3, to enhance hardware security without compromising TCAM functionality. Utilizing the existing MRAM-TCAM storage array, the design incorporates two $n : 1$ multiplexers (MUXs) and a cross-coupled sense amplifier (SA) for PUF response generation. In standard TCAM mode, the aforementioned auxiliary circuits remain deactivated to prevent any adverse impact on search operations. To achieve this, MUX selection signals are held disabled, maintaining outputs in a high-impedance (Hi-Z) state to electrically isolate the SA from the critical search data path. In PUF mode, the system executes the following sequence: First, the ML network is precharged. Then, an external challenge signal selects two specific target cells via the MUXs. Subsequently, SL and SLB are activated simultaneously, turning on the access transistors of all 2T-2MTJ cells in the selected rows. The inherent anti-parallel (AP) and parallel (P) magnetic tunnel junctions form parallel discharge paths, causing the ML voltage to decay. Finally, the cross-coupled SA senses the discharge rate difference (I_{cell1} vs. I_{cell2}) between the two target MLs and resolves this difference into a stable binary PUF response.

The entropy source of the proposed PUF originates from intrinsic, uncontrollable random physical variations in STT-MRAM manufacturing processes, effectively converted into measurable electrical parameter differences (specifically, discharge rate disparity). The physical foundation lies in the sensitivity of MTJ resistance to key process parameter fluctuations. Based on the quantum mechanical tunnel magnetoresistance effect (TMR), the MTJ resistance is critically determined by the oxide layer thickness (t_{ox}), energy barrier height (ϕ), barrier decay constant (κ), and the junction area (S). The P-state resistance R_P and AP-state resistance R_{AP} can be modeled as:

$$R_{P0} = \alpha \cdot \frac{t_{\text{ox}}}{S\sqrt{\phi}} \exp(\kappa \cdot t_{\text{ox}}\sqrt{\phi}) \quad (1)$$

$$R_{AP0} = R_P \cdot (1 + \text{TMR}) \quad (2)$$

where α is a material-dependent constant, and TMR is the tunnel magnetoresistance ratio. The access transistor (NMOS) operates in the linear region during the initial discharge, and its on-resistance is given by:

$$R_{DS0} = \frac{1}{\mu_n C_{\text{ox}} \frac{W}{L} (V_{\text{GS}} - V_{\text{th}})} \quad (3)$$

Here, μ_n represents electron mobility, C_{ox} is the gate oxide capacitance per unit area, W/L is the transistor aspect ratio, and V_{th} is the threshold voltage. Incorporating process variations, the resistances follow Gaussian distributions:

$$R_P = R_{P0}(1 + \Delta_P), \quad \Delta_P \sim \mathcal{N}(0, \sigma_P^2) \quad (4)$$

$$R_{AP} = R_{AP0}(1 + \Delta_{AP}), \quad \Delta_{AP} \sim \mathcal{N}(0, \sigma_{AP}^2) \quad (5)$$

$$R_{DS} = R_{DS0}(1 + \Delta_{DS}), \quad \Delta_{DS} \sim \mathcal{N}(0, \sigma_{DS}^2) \quad (6)$$

The total 2T-2MTJ cell resistance model comprises the parallel combination of the P-state branch ($R_P + R_{DS}$) and the AP-state branch ($R_{AP} + R_{DS}$):

$$R_{\text{unit}} = \frac{(R_{AP} + R_{DS})(R_P + R_{DS})}{R_{AP} + R_P + 2R_{DS}} \quad (7)$$

At $t = 0$, the match line (ML) is precharged to V_{DD} . Upon activation, the ML node voltage decays exponentially based on the total load capacitance C_{ML} :

$$V_{\text{ML}}(t) = V_{\text{DD}} \exp\left(-\frac{t}{R_{\text{unit}} C_{\text{ML}}}\right) \quad (8)$$

Due to process variations, $R_{\text{unit1}} \neq R_{\text{unit2}}$, leading to different discharge rates. This generates a transient voltage difference $\Delta V(t) = |V_{\text{ML1}}(t) - V_{\text{ML2}}(t)|$. The cross-coupled Sense Amplifier (SA) amplifies this difference. The initial rate of differential voltage growth, driven by the conductance mismatch, is derived as:

$$\frac{d|\Delta V(t)|}{dt} \approx \frac{V_{\text{DD}}}{C_{\text{ML}}} \left| \frac{1}{R_{\text{unit2}}} - \frac{1}{R_{\text{unit1}}} \right| \quad (9)$$

This initial seed voltage is subsequently latched by the positive feedback of the SA. The final PUF output bit is determined by the resistance polarity:

$$\text{Response} = \begin{cases} 1, & \text{if } R_{\text{unit1}} > R_{\text{unit2}} \\ 0, & \text{if } R_{\text{unit1}} < R_{\text{unit2}} \end{cases} \quad (10)$$

Furthermore, the 2T-2MTJ MRAM-TCAM cell within a column is configured as a data bit and a check bit pair: the control transistor gate of the data bit connects to the SL, while that of the check bit connects to the complementary SLB. A potential tampering or flip risk is flagged only when both MTJs are in the P state or both in the AP state, which corresponds to identical magnetoresistance states. In such cases, the cell output is forced to a mismatch state, flagging unreliable data. Conversely, if the MTJ states are complementary, the cell operates normally. This mechanism provides bit-level integrity verification for the TCAM array without requiring external security engine intervention.

Table 1
MTJ parameters and variables.

Parameter	Description	Value
TMR	Tunnel magnetoresistance ratio	2
t_{ox}	Oxide barrier thickness	0.85 nm
Dev-TMR	TMR variation percentage	3%
Dev-tsl	Switching layer thickness variation	3%
Dev-tox	Oxide thickness variation	3%
STO-dev	Switching duration variation	3%
RV	Gaussian variation factor	2
STO	Switching duration distribution	2
RA	Resistance-area product	5 $\Omega \mu m^2$
a	Long axis length	32 nm
b	Short axis length	32 nm

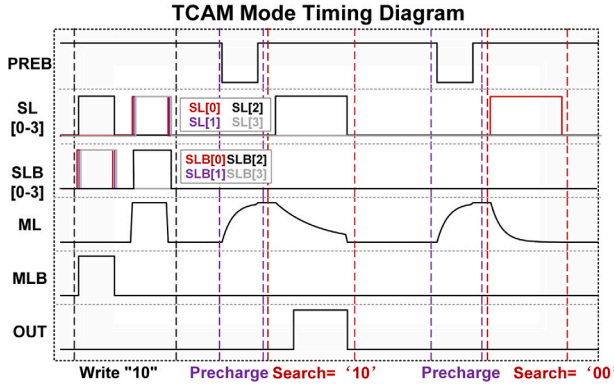


Fig. 4. Functional timing diagram of TCAM.

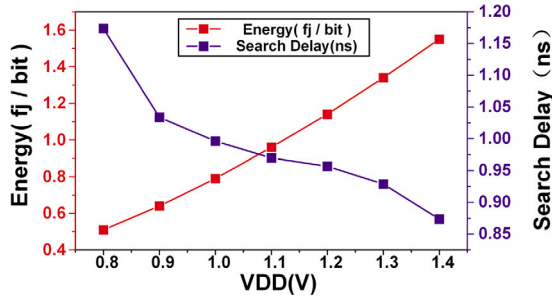


Fig. 5. Power consumption and search delay of 2-bit encoded MRAM-TCAM.

3. Experimental results and analysis

To systematically validate the key characteristics of the proposed dual-mode TCAM with integrated PUF design, this study employs a 65 nm CMOS platform combined with compact STT-MRAM device models. Table 1 details key simulation parameters. Monte Carlo statistical analysis using Cadence Virtuoso thoroughly examines both the energy efficiency and search performance of the TCAM design, as well as the PUF's core metrics including randomness, uniqueness, and reliability.

3.1. TCAM performance analysis

To validate the functionality of the proposed 2-bit encoded TCAM, comprehensive timing simulations were conducted. Fig. 4 illustrates the key signal timing for the complete 'write-precharge-search' cycle. The simulation begins with the data writing phase, where the precharge signal PREB is held high to ensure data stability. Following the proposed 2-bit encoding rule, the 4-bit mask '0100' (corresponding to pattern '10') is written into the target unit. Upon initialization, the system transitions to the search phase. This phase starts with a negative PREB pulse,

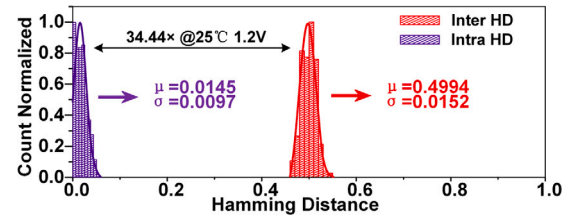


Fig. 6. PUF response uniqueness and native reliability.

precharging all MLs. Once PREB returns high, the search operation is initiated. When the search key '10' is applied, the encoding circuit generates the mask '0100' and activates SL1. Since the corresponding MTJ is in a high-resistance state, the ML retains a high voltage level due to negligible leakage current, producing a high OUT signal indicating a match. Conversely, applying search key '00' (encoded as '0001') activates SL0. The MTJ in this path, being in a low-resistance state, creates a rapid discharge path, causing the ML voltage to drop sharply and triggering a mismatch signal at the output. The transition between TCAM and PUF modes is completed within 1–2 clock cycles. To ensure data integrity, a mandatory precharge phase is enforced during the switch via the PREB signal, which clears residual charges on the match lines and prevents unintended disturbances to the 2T-2MTJ states. Furthermore, the 2-bit encoding logic is implemented in the periphery with minimal gates, incurring negligible routing overhead relative to the total array area.

Fig. 5 presents the measured per-bit search energy and search delay of the proposed TCAM design across the supply voltage range of 0.8 V to 1.4 V. The experimental data demonstrate that the search energy consumption maintains within the range of 0.51–1.55 fJ/bit throughout the entire operating voltage window. Notably, even at the maximum operating voltage of 1.4 V, the energy consumption remains lower than most comparative schemes operating at their nominal voltages. Concurrently, the search delay is maintained within 0.87–1.17 ns, demonstrating the design's low-latency characteristics.

Table 2 provides a detailed comparative analysis with representative TCAM schemes in recent literature. Experimental results based on the 65 nm technology node show that compared to the conventional 2T-2MTJ parallel TCAM design, the proposed 2-bit encoding scheme reduces search energy consumption from 2.72 fJ/bit to 1.08 fJ/bit at 1.2 V operating voltage, representing a 60.3% reduction. The corresponding search delay is 0.95 ns, achieving a 42.1% improvement over the unoptimized scheme's 1.64 ns. While the 2-bit encoding scheme does not reduce the intrinsic write energy of STT-MRAM, it strategically optimizes the system's holistic energy profile. In search-dominant TCAM applications, the 60.3% reduction in dynamic search energy — achieved by halving the match-line activation probability — rapidly offsets the initial write overhead, yielding superior long-term energy efficiency.

3.2. PUF performance analysis

PUF uniqueness quantifies the distinguishability between different instances, measured through Hamming Distance (HD). To quantitatively evaluate the uniqueness of the MRAM-TCAM-PUF, this study comprehensively investigated the inter-chip and intra-chip HD. Fig. 6 shows the HD distribution statistics for 25 PUF samples. The results indicate that the average inter-chip HD is 49.94%, with a distribution approximately normal, and the worst-case HD value is 46.39%. The average intra-chip HD is 1.45%. This comparison result confirms that the proposed PUF structure possesses excellent instance uniqueness and native reliability.

Autocorrelation function measures the dependency between response bits at different positions, where near-zero values indicate ideal

Table 2
TCAM performance comparison.

	JETCAS'23 [11]	TMA'25 [21]	TED'22 [22]	JJAP'10 [23]	Unoptimized	This work
Technology (nm)	28	7	40	90	65	65
Scale	32 × 32	64 × 128	—	144 × 256	32 × 32	32 × 32
Device	DMTJ	SOT-MTJ	Ge-MD	STT-MTJ	STT-MTJ	STT-MTJ
Search Energy (fJ/bit)	1.73	44	1.0	23.3	2.72	1.08
Search Delay (ns)	1.44	7.38	0.2	8	1.64	0.95

Table 3
NIST SP 800-22 randomness test.

Test item	Length	Proportion	P-value	Pass (%)
Frequency	2560	10/10	0.213309	100
Block Frequency	2560	10/10	0.350485	100
Cumulative Sums	2560	10/10	0.534146	100
Longest Run	2560	10/10	0.122325	100
Rank	2560	10/10	0.002043	100
FFT	2560	10/10	0.350485	100
Runs	2560	10/10	0.350485	100
Non Overlapping Template	2560	10/10	0.534146	100
Serial	2560	10/10	0.534146	100
Linear Complexity	2560	10/10	0.002043	100

randomness. Fig. 7(a) presents the autocorrelation analysis results of a total of 25,600 response bits, with 95% white noise confidence interval boundaries of 0.0125 and −0.0125, respectively. The analysis shows that the inter-bit correlation is weak, at the level of random noise, further supporting its uniqueness characteristics. Uniformity characterizes the balance of '0' and '1' bits in PUF responses, with optimal value at 50%. Fig. 7(b) shows the logical value distribution proportions of response bits in two representative datasets (labeled Scheme A and Scheme B). The results demonstrate that the PUF exhibits good uniformity in both configurations, with the proportions of '0' and '1' both close to 50%. Randomness evaluation employs statistical test suites to verify the unpredictability of PUF responses. The NIST SP 800-22 test suite provides comprehensive assessment through multiple statistical tests. We divided the total sample of 25600 bits into 10 segments of 2560 bits each, and applied the NIST SP 800-22 statistical test suite for multi-dimensional testing. The test results are shown in Table 3, where the MRAM-TCAM PUF successfully passed 10 selected tests.

Reliability measures the stability of PUF responses under varying environmental conditions, quantified by BER. Under extreme conditions with voltage increased to 1.4 V, the worst observed BER is 0.29%. Fig. 8(a) depicts the BER variation with temperature curve, where within the wide temperature range of −40 °C to 120 °C, the worst BER occurs at 120 °C, reaching 1.17%. Fig. 8(b) shows the BER trend with voltage variation. Quantitative analysis indicates that the average sensitivity coefficient of the PUF to temperature change is 0.091% per 10 °C, and to voltage change is 0.098% per 0.1 V, which are improved by 3.08 times and 29.39 times, respectively, compared to traditional TCAM PUF.

In hardware overhead, the PUF module consists of 267 transistors, including a 32-to-1 MUX (258 transistors) and a cross-coupled SA (9 transistors). Given the total transistor count of 5124 for the 32 × 32 MRAM-TCAM storage and peripheral base, the additional hardware overhead is accurately calculated as 5.21%. This minimal overhead justifies the integration of security features in resource-constrained environments.

Table 4 compares this work with recent representative PUF works in terms of process node, operating temperature, and voltage ranges. The comparison highlights three distinct competitive advantages. First, regarding environmental robustness, this work exhibits the highest immunity to environmental fluctuations. Within the extensive operating conditions of −40 °C to 120 °C and 1.0–1.4 V, the proposed design maintains the lowest BER sensitivity coefficients to both temperature and voltage variations among the compared work, ensuring

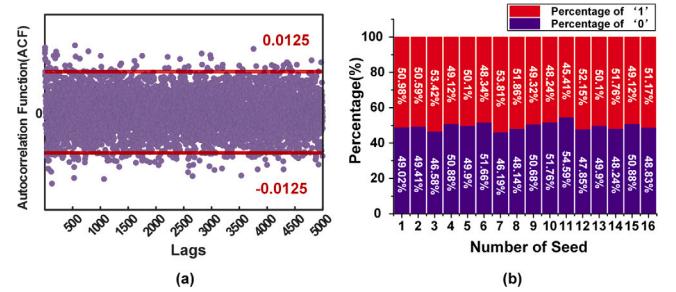


Fig. 7. (a) PUF response autocorrelation, (b) PUF response uniformity.

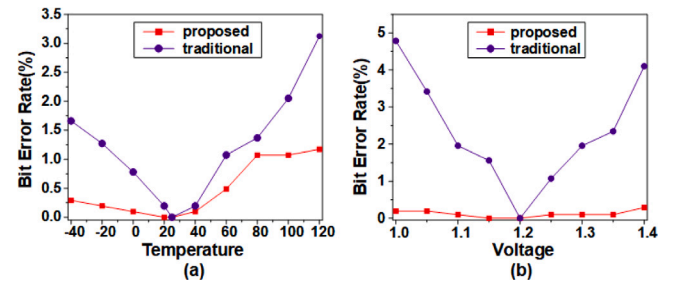


Fig. 8. PUF reliability performance: (a) BER under temperature variations, (b) BER under supply voltage variations.

consistent reliability in unstable edge environments. Second, the design achieves high energy efficiency; the measured bit energy of 1.08 fJ/bit is significantly lower than most reported MRAM/RRAM-based PUFs, offering a superior power budget. Finally, distinct from dedicated PUF macros, this work features a dual-mode architecture. By embedding the entropy source directly within the TCAM array, it enables intrinsic hardware authentication without incurring additional area overhead for the core array, providing a unique trade-off between security and resource utilization.

4. Conclusion

This paper proposes a dual-mode MRAM-TCAM with a 2-bit encoding scheme that reduces match-line activation probability, achieving a 60.3% reduction in dynamic search energy compared with conventional designs. In addition, an intrinsic PUF is integrated to enable hardware authentication by exploiting manufacturing variations, with only 5.21% hardware overhead. Simulation results demonstrate that the search energy consumption is only 39.7% of conventional designs, while the PUF maintains a worst-case BER of 1.17% at 120 °C and 0.29% at 1.4 V operating voltage, successfully passing NIST SP 800-22 randomness tests. This design provides a co-optimized solution for hardware security and energy efficiency in edge computing applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 4
PUF performance comparison.

	TC'22 [24]	TCE'23 [25]	SCI REP'24 [26]	ISCAS'21 [27]	VLSI'25 [28]	This work
Technology (nm)	–	45	28	28	40	65
Temp. Range (° C)	–20–90	–40–100	–25–100	–20–80	–25–125	–40–120
VDD Range (V)	0.18–0.22	0.8–1.2	0.65–0.85	0.5–1	0.5–1.1	1.0–1.4
BER per 10 °C (%)	0.39	1.37	0.34	0.76	0.25	0.091
BER per 0.1V (%)	7.8	0.2	1.75	0.09	0.435	0.098
Inter-HD (%)	50.64	49.99	49.96	51.1	49.94	49.94
Bit Energy (fJ/bit)	–	5.1	0.04/0.029	5.57	9.75	1.08
Dual-Mode	No	No	No	No	No	Yes

Data availability

No data was used for the research described in the article.

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